

Suggested hardware for the transceiver

From Radio Team (Santiago, Andrian, Marcos) **To** Carlos, Michele, Bikash, Guillem

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New in v0.3: all four teams moved to 2.4 GHz, so the radio is now **two Ebyte E01-ML01DP5** (nRF24L01+ with PA and LNA) per box, each feeding its own patch antenna. They need their own 3.3 V regulator. The Adafruit RFM69 breakout is no longer needed.

The radio baseline is 2.4 GHz with the nRF24L01+. Under our split, the electronics team picks the micro and designs the board. This page is the radio team's **suggestion**, built around every hardware rule in the competition PDF, to save you time. It is not a decision. Please challenge it.

In short

1. **Micro module: Raspberry Pi Pico 2** (RP2350, **not** the Pico 2 W). It costs about \$5, has 4 MB flash to hold the 1 MB file, 520 KB RAM and 3.3 V logic, and it can act as a USB host on any two pins or on its native USB port. It has no radio chip on board, so it is safe under the protocol ban.
2. **Radio: two Ebyte E01-ML01DP5 per box** (genuine nRF24L01+ with PA and LNA, SMA connector, 2 × 4 pin header), about \$4 each. One feeds the front patch antenna, one the back. They share one SPI bus. **VCC is 2.0–3.6 V only**, so they get their own 3.3 V regulator. The same modules go from the bench to the final boxes, because modules shown at Quick Mode are final. The radio team's driver is written and tested. (*Changed in v0.3: v0.2 used the Adafruit RFM69HCW breakout at 868 MHz.*)
3. **Board: a custom 2-layer carrier PCB** that the Pico 2 and the radio solder onto. It also carries the radio regulator, the USB-A host port, the battery charger, the 5 V supply for the stick, ESD protection, and the switch, button and LEDs. The two patch antennas are small separate boards on the box walls, fed by SMA pigtails. Before Quick Mode we use a perfboard prototype with the **same modules**.
4. **Power: one protected 18650 Li-ion cell**, charged on board through a charge-only USB-C port. Our whole competition day uses roughly 10–20 % of it.
5. **Estimated cost: about €45–60 per device** with a lab-made PCB, or about €35–45 with an external fab, against the €150 cap. If the lab counts each patch board as its

own €15 board, add up to €30; panelising the patches with the carrier may avoid that (ask the lab).

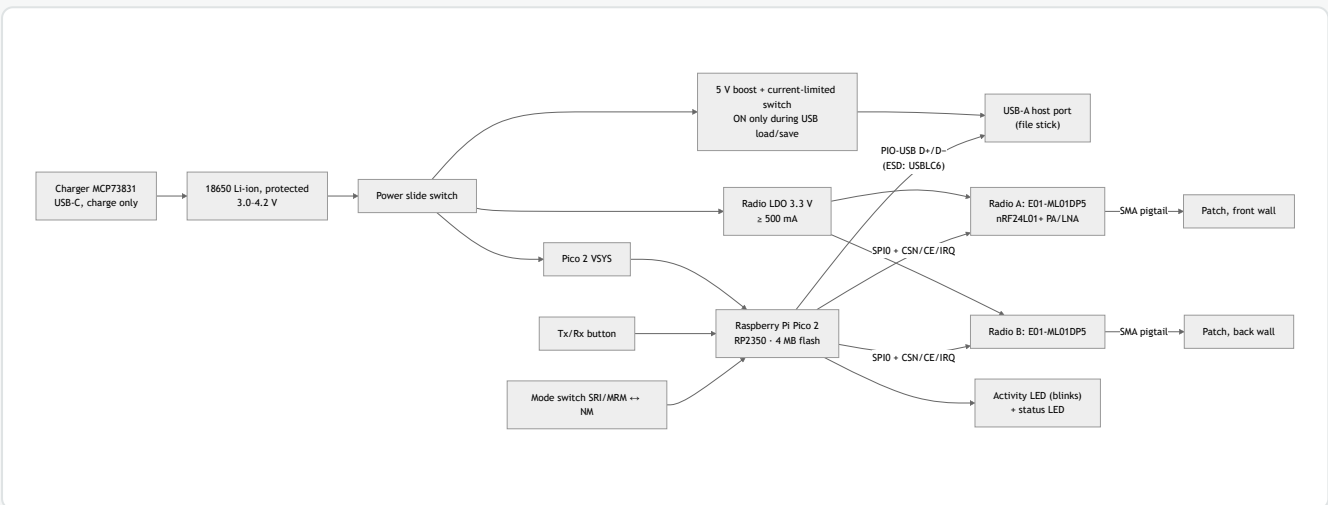
The rules this design answers

RULE (COMPETITION PDF)	WHAT IT FORCES ON THE BOARD
One and only one module including a microprocessor; no single commercial module embedding the entire system	Pico 2 is the only micro. The radio, charger and boost contain no CPU. No USB-host chips that have their own controller (e.g. CH376), which could count as a second micro.
No chip implementing WiFi, Bluetooth, ZigBee, 2G–5G	Banned even if unused: Pico W / Pico 2 W, ESP32 boards, Arduino Nano 33 BLE/IoT, Uno R4 WiFi, Nano ESP32, Raspberry Pi Zero W / 3 / 4 / 5.
Read/write the file from/to a USB memory device: 5 min setup while still wrapped in foil, 3 min to upload afterwards	A USB-A host port on the box wall, plus ≥ 1 MB on-board storage . The sender copies the file in at setup, and the receiver keeps what it got until upload. The stick can be removed during the round.
Fits entirely in the 15 × 15 × 7 cm IKEA 365+; only connectors, switches and buttons may stick out	PCB about 100 × 80 mm or smaller. The antennas sit inside: two ~60 × 60 mm patch boards standing on the front and back walls, as high as possible.
Operates autonomously	A battery and a power switch. Nothing plugged in during a round.
At least one Tx/Rx switch or button; a pilot light that blinks in Tx/Rx	Panel pushbutton for Tx/Rx and an activity LED. The box is transparent, so the LED can stay inside.
Extra switches allowed (e.g. network mode), but no switch indicating the position in the network	One optional SRI/MRM ↔ NM mode switch. Nothing that encodes a node number.
Drop test from table height to concrete (before MRM)	Everything soldered (no sockets or loose headers), board on standoffs, battery strapped, antenna glued down.
ESD and RF exposure compliance	TVS/ESD protection on USB and the buttons. RF power is small (≤ 10 mW e.i.r.p.).
Modules shown at Quick Mode are final	The QM prototype uses the same Pico 2 and the same E01-ML01DP5 modules as the final board. Antennas may differ (a whip at QM).

RULE (COMPETITION PDF)	WHAT IT FORCES ON THE BOARD
Unit cost $\leq$ €150; lab-made PCB counts €15; everything (screws, foam, wires) goes in the BoM	Cheap modules; consider an external fab for the PCB (see the cost table).

Block diagram

The only fast data path is SPI to the radios, at 31 kB/s (250 kbps). USB file transfer happens only before and after a round.



Micro module options

Ranked against what the board must do: hold a 1 MB file, act as a USB host, use 3.3 V logic, carry no radio chip, stay cheap, and be ready before Quick Mode.

MODULE	STORAGE FOR 1 MB FILE	USB HOST	PRICE	VERDICT	NOTES
Raspberry Pi Pico 2 (RP2350)	4 MB flash ✓	PIO-USB on 2 GPIOs, or native USB	~\$5	recommended	Native USB stays free for programming and logs. Big community. Castellated edge, so it solders flat (good for the drop test).

MODULE	STORAGE FOR 1 MB FILE	USB HOST	PRICE	VERDICT	NOTES
Raspberry Pi Pico (RP2040)	2 MB flash, tight	Same as above	~\$4	fallback	Works, but 1 MB of file plus firmware leaves little headroom.
Adafruit Feather RP2040 USB Host	8 MB flash ✓	USB-A on board	~\$15	fastest prototype	Less to build, but costs more, and we would show less electronics design of our own (which is graded).
Teensy 4.1	8 MB flash ✓	Host port + library	~\$35	expensive	Easiest USB host, but it hurts the cost category.
STM32F411 "Black Pill"	512 KB ✗	Native OTG	~\$5	needs extra flash	Would need external flash, and USB mass-storage host firmware is heavier.
Arduino Uno / Nano (AVR)	✗	✗	—	no	5 V logic (the radio is 3.3 V only), 2 KB RAM, no USB host.
Raspberry Pi Zero (non-W)	SD card ✓	✓	~\$5–10	no	Linux: slow boot, high current, and SD corruption on power loss or a drop.

MODULE	STORAGE FOR 1 MB FILE	USB HOST	PRICE	VERDICT	NOTES
Pico W / Pico 2 W, ESP32, Nano 33 BLE/IoT, Uno R4 WiFi, Pi Zero W, Pi 3/4/5	—	—	—	banned	On-board WiFi/BT chip.
Micro + radio combos (Feather RP2040 RFM69, Moteino)	—	—	—	avoid	Risks the "single commercial module embedding the entire system" rule. Would need an Atenea answer first.

Prices are list prices seen online on 7 Oct 2026 (Raspberry Pi, TME, Adafruit). Get real quotes before ordering.

Suggested parts per device

Example part numbers to start from. Electronics owns the final choice, the footprints and the sourcing.

FUNCTION	SUGGESTED PART	WHY	≈ € / DEVICE
Micro module	Raspberry Pi Pico 2 (SC1631), non-W	See above	5.5
Radio modules	2 × Ebyte E01-ML01DP5 (nRF24L01P + PA/LNA, SMA), on 2 × 4 headers. Not E01C-* (clone chip)	Breadboard-ready, chip-controlled PA/LNA, so only SPI + CSN + CE + IRQ. Same modules from the bench to the final box	7.5
Antennas	2 FR4 patch boards (~60 × 60 mm, front and back walls), 2 SMA pigtails, a 0.5–3 dB SMA pad if T0 needs it; 2 dBi SMA whip for Quick Mode	Vertical polarisation, mounted high. The pad trims the output to exactly 10 dBm e.i.r.p.	6 + boards

FUNCTION	SUGGESTED PART	WHY	≈ € / DEVICE
USB host port	USB-A receptacle, through-hole, panel-facing	Through-hole survives plugging and the drop test	0.5
5 V for the stick	Boost (e.g. TPS61023) + current-limited switch (e.g. TPS2051 / AP2552), enabled by a GPIO	Sticks need 5 V. Powered only during load/save, which saves energy and protects against shorts	2
ESD	USBL6-2SC6 on D+/D-; TVS on VBUS and the button lines	ESD compliance; the USB port is touched every round	0.6
Battery	1 × protected 18650 Li-ion (~2500 mAh) with solder tabs, or holder + strap	Steel can is robust in the drop test; rechargeable (sustainability report)	6–8
Charger	MCP73831 (or TP4056) + USB-C receptacle, charge only	Charge between rounds without opening the box. Nothing is plugged in during operation	1.2
Controls	Slide power switch, panel Tx/Rx pushbutton, optional mode switch, 2 LEDs	Required by the rules	2.5
Radio supply	3.3 V LDO (AP2112K-3.3 or TLV75533) + 100 µF / 10 µF / 100 nF at each module; 100 µF on USB VBUS	E01 VCC is 2.0–3.6 V; PA/LNA modules are sensitive to supply noise; handles ~130 mA TX bursts and stick inrush	1
PCB	2-layer carrier, ≤ 100 × 80 mm	ETSETB lab = €15 in the BoM (rule); an external fab is ~€1 per board but adds 1–2 weeks	1–15
Mechanics	M3 nylon standoffs, base plate, cable ties, hot glue, IKEA 365+ box	Drop test; the BoM must list them too	5–7
Total per device (verify with quotes; + up to €30 if each patch board counts as a lab board)			≈ 35–60

Power budget

Current draw

Pico 2 running	≈ 30–50 mA
	ESTIMATE
E01 TX (one at a time)	≈ 130 mA peak
	ESTIMATE
E01 RX, both listening	≈ 2 × 26 mA
	DATASHEET
USB stick (5 V)	50–100 mA, only during load/save
	ESTIMATE
Peak on the radio LDO	≈ 160 mA (radios no longer load the Pico's 3V3)

Battery sizing

A sending round averages about 170 mA. Idle with both radios listening is about 100 mA. A full competition day of ~4 hours powered on needs about **400–700 mAh**. One 2500 mAh 18650 covers that about four times over. A smaller LiPo would also do, but pouch cells are more fragile in the drop test.

For the sustainability report

We can measure watts per round directly on this design. The parts are rechargeable and RoHS. The modules unsolder and can be reused, and the board shape lets you repair rather than replace.

Build stages

STAGE	BOARD	GOAL	TARGET
1 · Bench	Pico 2 + E01-ML01DP5 with a whip on a breadboard, USB powered	Radio tests T0 (output power) and T2; USB-stick read/write via PIO-USB	as soon as parts arrive
2 · Quick Mode	Perfboard, same module part numbers , external power allowed	One error-free line from a file in < 2 min, first attempt	week 6–7
3 · PCB rev A	Custom 2-layer carrier (KiCad), boxed, battery	Autonomous, drop-tested, range tests T4/T5	week 8–9
4 · PCB rev B	Fixes from rev A	Pretest (4 Dec) fully packaged	week 10–11

Risks and how we cover them

RISK	MITIGATION
PIO-USB host fails with some USB sticks	Test 3–4 different sticks in stage 1. The RP2350's native USB can also be a host, so a second path exists on the same chip.
The foil-wrapped setup makes the USB port hard to reach	Put the port on a side wall. Ask on Atenea whether the foil may be opened at the port during the 5-min setup.
Brown-out or noise during radio TX	Separate radio LDO and bulk capacitance at each module; check the battery at low charge (3.3 V) in stage 3.
Clone radio chips (Si24R1)	Buy only the plain E01-ML01DP5 part number from Ebyte's store or a distributor; T0/T2 check the chip.
Patch antenna off frequency (FR4 tolerance)	A tuning coupon with five lengths, measured on the RF-lab VNA inside the box before the final patch boards.
Drop test breaks a joint or ejects the battery	No sockets; standoffs; strapped cell; drop rev A on purpose in week 9.
The "attacks" interrupt the link mid-round	Watchdog and brown-out reset; firmware resumes from the last acknowledged line (protocols team).

What we need from electronics

1. Agree or counter-propose the micro by **Fri 16 Oct**, so we can order together with the radio modules.
2. Confirm the E01-ML01DP5 and the radio LDO, and pick the PCB route (lab at €15, or external fab). Ask the lab whether the patch boards can be panelised with the carrier as one board.
3. Pin map for SPI and the two radios' CSN, CE and IRQ (proposal: GP16/18/19; A on GP17/20/21; B on GP22/14/15), PIO-USB D+/D–, button, LEDs and mode switch. We will write the radio's hardware layer against it.
4. The IKEA box's inside height: a patch board needs about 62 mm.
5. Book Prof. Chávez or Prof. Jiménez to review this block diagram and the lab's PCB capabilities.

Sources checked 7 Oct 2026: [Pico-PIO-USB](#) (USB host/device via PIO, RP2040 and RP2350) · [Pico 2 specs](#) · [Ebyte E01-ML01DP5](#) · [nRF24L01+ Product Specification](#).